Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Shoukas, Artin A. Ph.D.		
Project Title:	Non-Adrenergic Mechanisms of Cardiovascular Deconditioning		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Baltimore	**State:**	MD
Zip Code:	21205-2109	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	03/01/2004	**End Date:**	02/29/2008
No. of Post Docs:	3	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Berkowitz, Dan (The Johns Hopkins University) Hare, Joshua (The Johns Hopkins University) Nyhan, Daniel (The Johns Hopkins University) Li, Dechun (Johns Hopkins Medical School)		
Grant/Contract No.:	NCC 9-58-CA00405		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Cardiovascular remodeling in response to microgravity predisposes astronauts to orthostatic intolerance and decreased exercise tolerance on return to a 1G environment. It is now appreciated that altered loading of the heart and blood vessels, and neuro-hormonal signals, drive these changes which result in remodeling and cardiovascular de-conditioning. While central autonomic processing and efferent sympathetic and parasympathetic responses have been shown to contribute to orthostatic intolerance, end organ hypo-responsiveness has recently been shown to be of critical importance in the de-conditioning/hypo-responsive phenotype. In new data within this grant, we demonstrate that nitric oxide, an important cardiovascular signaling molecule, is an important mediator of cardiovascular de-conditioning. We demonstrate: 1) the NO/cGMP pathway is up-regulated and ontributes to vascular hypo-responsiveness; 2) myocardial contractile reserve is depressed in cardiac myocytes from animal models of microgravity; 3) the presence of a beta adrenergic receptor (â-3AR) coupled to NOS-3 that acts as a negative modulator of â-AR mediated enhanced myocardial contractility; and 4) the critical role of the NO/cGMP system in the prevention of myocardial hypertrophy. We hypothesize that the NO/cGMP pathway is up-regulated in both the heart and vasculature resulting in vascular and myocardial contractile hypo-responsiveness, as well as vascular and myocardial atrophy. We will determine the role of NO/cGMP system in the vascular hypo-responsiveness, myocardial contractility and cardiac remodeling, as well as the interaction of vessels and heart, ventricular-vascular coupling. Furthermore, based on our findings, we propose novel countermeasures aimed at inhibiting these dysregulated pathways.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These findings could have important therapeutic significance for both astronauts and long-term bed rest patients. Both of these groups are predisposed to orthostatic intolerence upon returning to normal gravity and returning to the upright position respectively. These findings could also have important therapeutic significance for the normal aging population.
Task Progress:	1. The role of the endothelium and the NO/cGMP pathway in vascular hypo-responsiveness and remodeling in microgravity is nearly completed and will be finalized with two manuscripts mid year of year 2 of this proposal. . 2. The role of NO in modulating myocardial contractility and hypertrophy has just begun and we expect to finish these studies during year 2, this coming year, of the project. 3. Whether up-regulation of the NO/cGMP system results in decreased vascular and ventricular stiffness as measured by integrated cardiovascular function and ventricular-vascular coupling will begin toward the end of year 2 of this project. We have delayed working on this aspect of the project since it relies on using our chronic rat preparation. This preparation requires that we purchase micro vascular pressure telemetry system to monitor the rats on a 24 hour basis.
Bibliography Type:	Description: (Last Updated: 05/05/2011)
Articles in Peer-reviewed Journals	1. Barouch LA, Berkowitz DE, Harrison RW, O' Donnell CP, Hare JM "Disruption of leptin signaling contributes to cardiac hypertrophy independently of body weight in mice Circulation" AHA, 754-759 , Aug-2003
Articles in Peer-reviewed Journals	3. Khan SA, Lee KH, Minhas KM, Gonzalez DR, Raju SV Y, Tejani AD, Li D, Berkowitz DE, Hare JM. "Neuronal nitric oxide synthase negatively regulates xanthine oxidoreductase inhibition of cardiac excitation-contraction coupling. Proc Natl Acad Sci" 15944-15948 , Nov-2004
Articles in Peer-reviewed Journals	4. Jeon BH, Gupta G, Park YC, Qi B, Haile Z, Khanday FA, Liu Y- X, Kim JM, Ozaki M, White R, Berkowitz DE, Irani K "Apurinic/apyrmidinic endonuclease 1 regulates endothelial NO production and vascular tone. Circ Res" 902-910 , Jan-2005
Awards	" Nominated as Treasurer for the Biomedical Engineering Society " Jan-2005
Awards	" The Golden Apple Teaching Honor. Awarded by the Medical School Class of for outstanding professorship in teaching. " Jan-2005
Presentation	Shoukas, AA "Reflex Control in Simulated Micro Gravity " May-2005
Presentation	Shoukas, AA "Veins and Their Control " Apr-2004